

ITECA3-12 Initial Project

Site: [Eduvos LMS](#)
Course: Web Development and e-Commerce Assessments
Book: ITECA3-12 Initial Project

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Description

This project requires you to **plan, develop, and present** a **Consumer -to-Consumer (C2C) e-commerce system**. You will first propose your solution, then design and build a working system, and finally present it with supporting documentation

Table of contents

1. Project / Research Essay
2. Instructions to Students
3. Section A
4. Scenario
5. Deliverable 1
6. Deliverable 2
7. Deliverables 3

Project

Faculty:	Information Technology
Module Code:	ITECA3-12
Module Name:	Web Development and E-Commerce
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Total Marks:

100

Submission Week:

Deliverable 1: Block 1, Week 4

Deliverable 2: Block 2, Week 5

Deliverable 3: Block 2, according to summative assessment schedule

Project Resubmission:

This module is presented on NQF level 7.

5% will be deducted from the student's project mark for each calendar day the project is submitted late, up to a maximum of three calendar days. The penalty will be based on the official campus submission date.

Projects submitted later than three calendar days after the deadline or not submitted will get 0%. [\[1\]](#)

This is an individual project.

This project contributes 70% towards the final mark.

[1] Under no circumstances will projects be accepted for marking after the projects of other students have been marked and returned to the students.

Instructions to Students

1. Please ensure that your answer file (where applicable) is named as follows before submission: **Module Code – Assessment Type – Campus Name – Student Number.**
2. Remember to keep a copy of all submitted projects.
3. All work must be typed.
4. Please note that you will be evaluated on your writing skills in all your projects.
5. All work must be submitted through Turnitin. The full originality report will be automatically generated and available for the lecturer to assess. Negative marking will be applied if you are found guilty of plagiarism, poor writing skills, or if you have applied incorrect or insufficient referencing. (See the "instructions to students" book activity before this activity where the application of negative marking is explained.)
6. You are not allowed to offer your work for sale or to purchase the work of other students. This includes the use of professional assignment writers and websites, such as Essay Box. You are also not allowed to make use of artificial intelligence tools, such as ChatGPT, to create content and submit it as your own work. If this should happen, Eduvos reserves the right not to accept future submissions from you.

Section A

Learning Objective

This project focuses on key web technologies and tools for web application development and the essentials of e-commerce. Instead of giving students a specific project to replicate, students are given the freedom to decide on a project solution for the given scenario.

Students are also expected to conduct requirement analysis, gather all information related to the platform the aim to build, and formulate a project proposal. Thereafter, build a database-driven e-commerce website, utilizing HTML, CSS, JavaScript, PHP, and MySQL to meet the client's needs or requirements.

Project Topic: C-2-C e-Commerce platform development.

Scope: In this project, students are required to design and develop a C-2-C e-commerce platform used to facilitate the selling and buying of various goods between customers. Students must complete and submit each project deliverable using the provided template and develop the website using HTML, CSS, JavaScript, PHP, and MySQL.

Scenario

Study the scenario and complete the question that follow:

C-2-C E-Commerce and the South African Informal Economy

According to the **Online Retail in South Africa 2025** report, the country's e-commerce turnover is expected to exceed **R130 billion** by the end of 2025 (World Wide Worx & Mastercard, 2025). This growth is driven by a structural realignment where digital platforms are no longer an experimental force in the economy, capturing nearly 10% of total retail spend. While formal retail giants dominate the B-2-C space, a significant portion of this digital expansion is occurring within the informal and C-2-C sectors as individuals seek new ways to generate income.

The **Quarterly Labour Force Survey** highlights that the informal sector remains a critical lifeline, accounting for nearly **20% of total employment** in the country (Statistics South Africa, 2025). Many informal traders and "side-hustlers" are moving away from traditional street vending toward digital platforms to reach more customers. However, most participants still operate without formal business registration or dedicated digital infrastructure. This "Township Economy," valued at approximately **R900 billion annually**, presents a massive opportunity for domestic C-2-C platforms that bridge the gap between informal trade and secure, digitized marketplaces (Standard Bank, 2025).

Research published in the **Global Media Journal** indicates that while the adoption of online shopping among emerging township consumers is heavily influenced by consumer attitudes toward convenience and perceived usefulness (Makhitha & Ngobeni, 2023). Despite the popularity of social media trade, many users face significant hurdles, including a lack of localized trust in online payments and the absence of integrated logistics for non-urban areas. There is a clear need for locally developed platforms that offer secure transactions and verified seller identities to reduce digital fraud and build long-term consumer loyalty (World Wide Worx & Mastercard, 2025).

To boost the domestic economy, there is a strategic push to develop C-2-C infrastructure that keeps transaction fees and revenues within South Africa. By creating platforms specifically designed for the local context supporting multiple languages, low-data interfaces, and integration with local payment gateways, developers can help formalize the informal sector. This digital enablement allows small-scale sellers to move from "survival to scale" and achieve financial inclusion and providing a sustainable pathway to economic growth for millions of South Africans (Standard Bank, 2025).

Reference List

Global Media Journal (2023). *Factors Influencing the Online Clothing Shopping Intention of Emerging Township Consumers in South Africa: The Mediation Effect of Attitude.*

- URL: <https://www.globalmediajournal.com/open-access/factors-influencing-the-online-clothing-shopping-intention-of-emerging-township-consumers-in-south-africa-the-mediation-effect-of-attitude-aid=92394>

Standard Bank (2025). *Township Informal Economy Report (October 2025).*

- URL: https://www.standardbank.co.za/static_file/South%20Africa/PDF/Township/Standard_Bank_Township_Informal_Economy_Report_October_2025.pdf

Statistics South Africa (2025). *Quarterly Labour Force Survey (QLFS) – Q1: 2025.*

- URL: https://www.statssa.gov.za/?page_id=1854&PPN=P0211

World Wide Worx & Mastercard (2025). *South Africa's Online Retail Set to Surpass R130 Billion in 2025.*

- URL: <https://www.mastercard.com/news/eemea/en/newsroom/press-releases/en/2025-1/september/south-africa-s-online-retail-set-to-surpass-r130-billion-in-2025/>

Deliverable 1

30 Marks

Task Description: Research the current state of C-2-C e-commerce platforms in South Africa, focusing on how they can be used to improve the informal economy. Identify specific ways to develop a platform that enables secure and efficient C-2-C trade. Based on your research, write a full project proposal for the platform you intend to build using the template provided.

1.1 Introduction

Before offering the details of your proposal, you must demonstrate an understanding of your customer's background. This includes a description of how their current operations are being impacted, or a description of their desired future state.

1.2 Needs/Problems

Identify the needs or problems to be addressed. Include the target population and any statistical information that you may have. Ideas for information to include here are:

- Length of time needs/problems have existed
- Whether problem has ever been addressed before, and what the outcome was
- Impact of problem to target population
- Impact of problem to surrounding populations

1.3 Goals/Objectives

State the desired goals and objectives to address the needs/problems stated above. Also include key benefits of reaching goals/objectives.

- Specific & measurable goal 1
- Specific & measurable goal 2
- Specific & measurable goal 3 etc.

1.4 Procedures/Scope of Work

Provide detailed information about proposed procedures, if available, and the scope of work. Include information on activities such as training, testing, and actual work required.

1.5 Timetable

Provide detailed information on the expected timetable for the project. Break the project into deliverables and provide a schedule for each deliverable.

	Description of Work	Start and End Dates
Deliverable 1		
Deliverable 2		

Deliverable 3		
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Also use a Gantt chart for more detailed project timetable.

1.6 Conclusion

Mark Allocation:

Criteria	Description	Mark Allocated
1.1 Introduction	Background and overview of the chosen platform.	5
1.2 Needs/Problem	Detailed problem statement regarding the South African C-2-C landscape.	6
1.3 Goals/Objectives	Clear, measurable objectives for the proposed solution.	6
1.4 Procedures/Scope	High-level overview of the work to be performed.	5
1.5 Timetable	A realistic timeline for project completion.	3
1.6 Conclusion	Summary of the proposal's potential impact.	5
Total		30

Due Date: Block 1, Week 4

End of Deliverable 1

Deliverable 2

50 Marks

Design a prototype for the e-commerce platform specified in your Project Proposal. The platform must allow customers to buy or sell goods and include an administrative website for management. The admin site must support Role-Based Access Control (RBAC) to create, display, update, and delete different user types.

Required Diagrams:

- Class Responsibility Collaborator (CRC) Cards
- Enhanced Entity Relationship Diagram (EERD)
- Context Diagram
- Data Flow Diagram (DFD)
- Use Case Diagram
- Database Design (Schema)

Technical Requirements:

- **Languages:** HTML, CSS, JavaScript (jQuery accepted), PHP, MySQL.
- **Frameworks:** Bootstrap is permitted; however, CMS tools (WordPress, Wix, etc.) are **strictly prohibited**.
- **Hosting:** The application must be hosted online (No localhost submissions).
- Any other tools or web development technologies can also be explored and used, depending on the nature of your specific project.

Important Notes for Students

Note 1: Formal Submission The link to the completed and hosted website, the GitHub repository link, and/or the zipped folder containing your source code must be formally submitted and shared with your lecturer **ONE (1) week BEFORE** your Deliverable 3 presentation.

Note 2: No Localhost Permitted The submission of **localhost** is **NOT** permitted for the Deliverable 3 presentation. Failure to host your website on a live server will result in a **50% Presentation Mark penalty**.

2.1 Introduction

For the purpose of this project, all student submissions must be based on a Customer-to-Customer (C2C) e-commerce model. Students are required to design and develop an online platform that facilitates the exchange of goods and services directly between individual consumers. Alternative e-commerce models, such as Business-to-Consumer (B2C), Business-to-Business (B2B), or hybrid models, are not permitted for this assessment.

Provide an introduction in less than 200 words.

2.2 Prototyping

Provide the screen shots of the responsive prototypes asked in the project specification document. Each prototype must illustrate how all the pages of the website will be displayed on smart phones, tablets, and desktops.

- a. Main website.
- b. Admin website.

2.3 Designing

- a. Class Responsibility Collaborator (CRC) cards
- b. Enhanced Entity Relationship Diagram (EERD)
- c. Context Diagram
- d. Data Flow Diagram (DFD)
- e. Use Case Diagram

2.4 Coding

Provide the screenshots and code samples of the final website. You do not have to print every code on your website. Use the following as guideline and explain what each sample code provided is used for on the website:

- a. Screenshots
- b. Sample PHP Code
- c. Sample HTML Code
- d. Sample JavaScript Code
- e. Sample CSS Code
- f. Sample MySQL Table Screenshots

2.5 Conclusion

Mark Allocation:

Criteria	Description	Mark Allocated
Introduction	Overview of the prototype and technical stack.	2
2.1 Prototype	C-2-C platform	6
	Organization admin website	6
2.2 Design Diagrams	Class Responsibility Collaborator (CRC) cards	3
	Enhanced Entity Relationship Diagram (EERD)	3

	Context Diagram	3
	Data Flow Diagram (DFD)	3
	Use Case Diagram	3
	Database Design	3
2.3 Coding	Platform Screenshots	2
	Sample PHP Code	3
	Sample HTML Code	3
	Sample JavaScript Code	3
	Sample CSS Code	3
	Sample MySQL Table Screenshots	2
2.4 Conclusion	Summary of the design and development phase.	2
Total		50

Due Date: Block 2, Week 5

End of Deliverable 2

Deliverables 3

20 Marks

You are required to present your final project (the complete platform) to your lecturer. The web applications must be hosted using a live hosting provider (e.g., InfinityFree, 000webhost, or similar free tiers). Along with the presentation, you must submit a comprehensive **User Manual** designed to instruct users on how to utilize the web applications effectively.

The User Manual must include:

- **Technical Stack:** A detailed list of the technologies used (e.g., PHP, MySQL, JavaScript).
- **System Features:** A breakdown of the platform's core functionalities for both customers and administrators.
- **Operational Guide:** Step-by-step instructions on how to use each feature, supported by screenshots.

Mark Allocation:

Criteria	Description	Mark Allocated
3.1 User Manual	Evaluation of the manual's clarity, technical stack documentation, and step-by-step feature guides.	5
3.2 Presentation	Live demonstration of the hosted C2C platform, including a walkthrough of the admin and user interfaces.	15
Total		20

Due date: Block 2, According to Summative Assessment Schedule.

End of Deliverable 3